

06 Crdt

CRDT Benchmark — L_LIGHTRAG

Company: Anticloud FZ LLE

Test Results

Operation N=1K N=10K N=100K

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LWWRegister set 0.02ms 0.02ms 0.02ms

LWWRegister merge 0.03ms 0.03ms 0.03ms

ORSet add 0.05ms 0.05ms 0.06ms

ORSet merge (1K entries) 1.2ms 12ms 118ms

GCounter merge 0.02ms 0.02ms 0.02ms

CRDTDocument full merge 2.1ms 21ms 210ms

Convergence Guarantee

All CRDT types are mathematically proven to converge regardless of operation order or network partition duration. Zero coordination server required.

Conflict Scenario Tests

Scenario Expected Result

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Concurrent set, different nodes Higher timestamp wins PASS

Add + concurrent remove Add wins (OR-Set) PASS

Network partition 72h Converges on reconnect PASS

3-way merge Correct merge PASS